

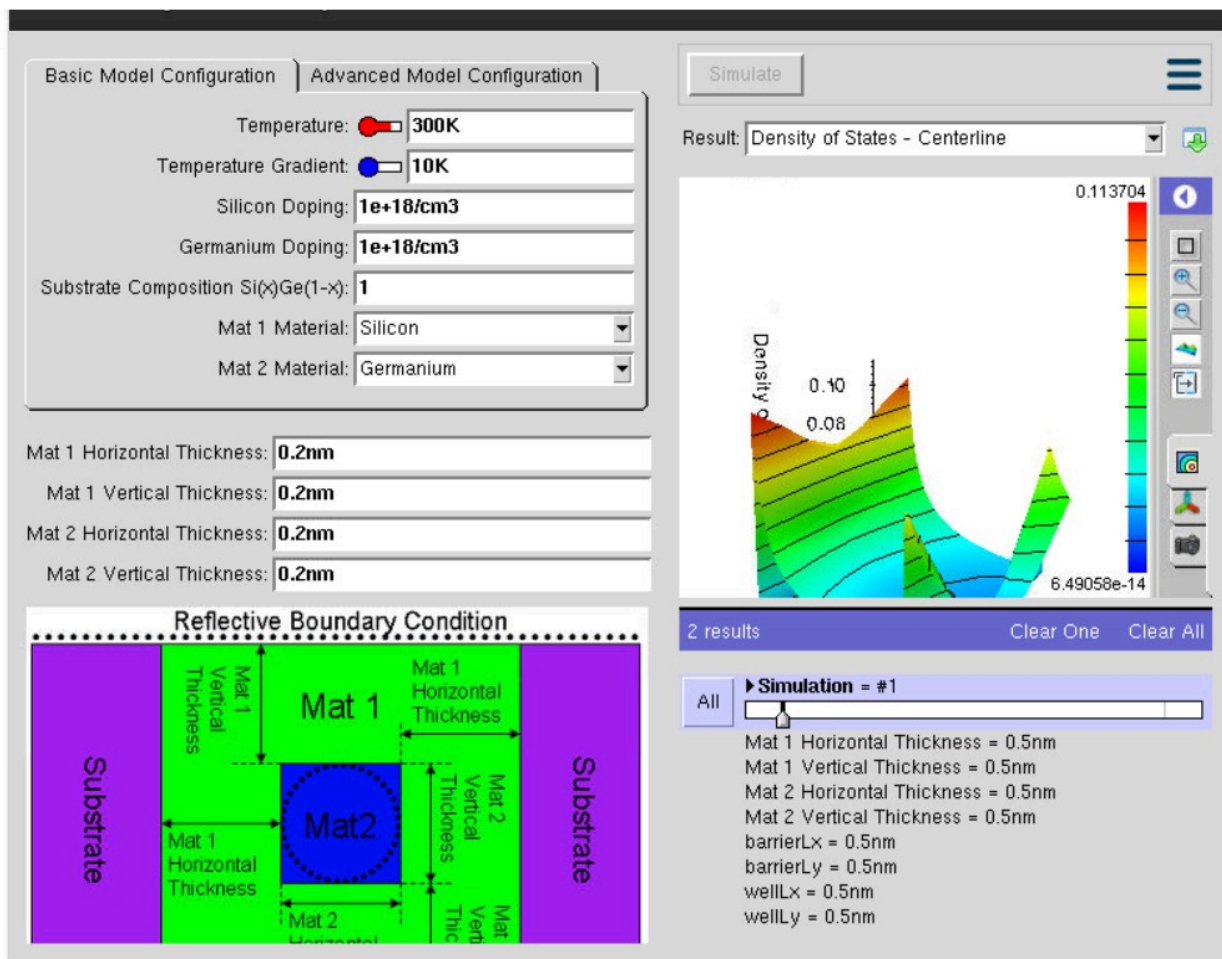
Exp. 2. Simulation of Density of States of Quantum Well Structure with Various Nanometer Thickness.

Casestudy

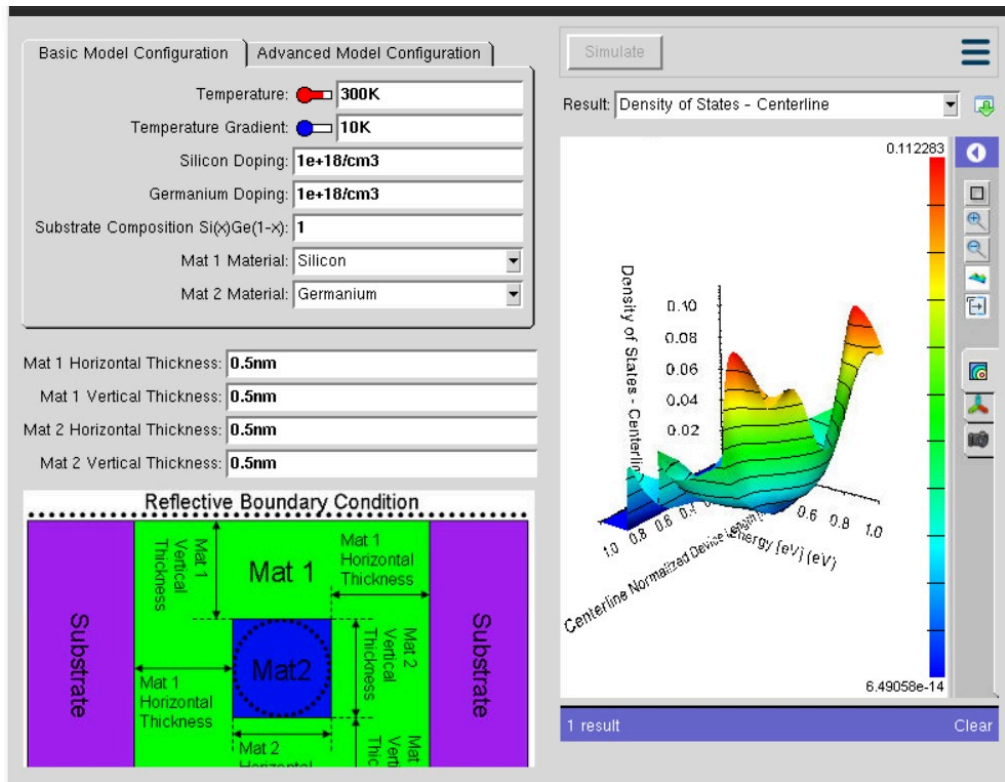
Case Study 1: Simulate the density of states of quantum well of thickness 0.2 nm



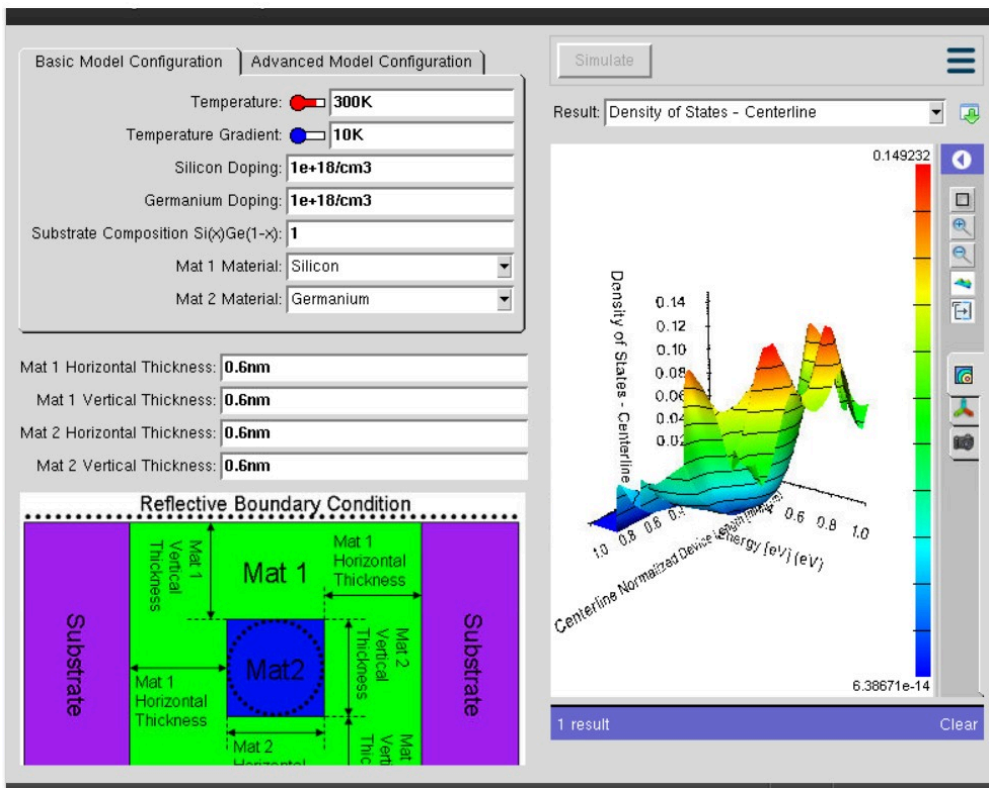
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Case Study 2: Simulate the density of states of quantum well of thickness 0.5 nm

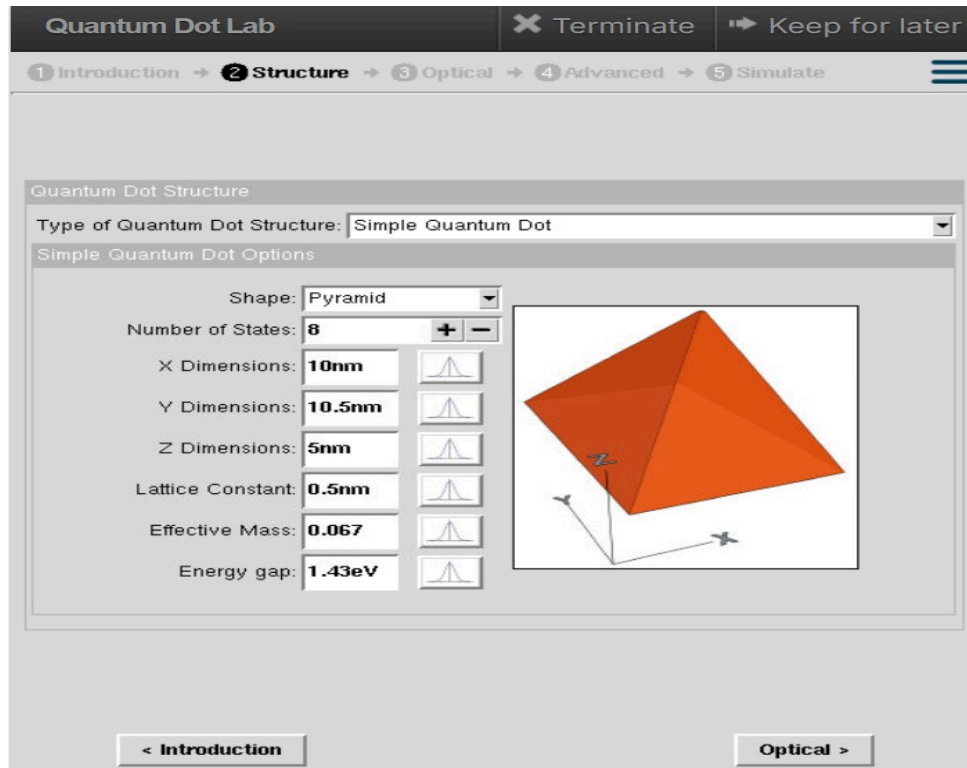


Case Study 3: Simulate the density of states quantum well of thickness 0.6 nm



Exp. 3. Simulation of Energy States of Quantum Dot using NanoHub

Case Study 1: Simulate the energy states of Triangle quantum dots of $X=10\text{ nm}$, $Y = 10.5\text{nm}$ and $Z = 5\text{ nm}$.



Quantum Dot Lab

✕ Terminate

➡ Keep for later

1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate

Light Incident Angles

Angle Theta: 45deg

Angle Phi: 0deg

Sweep

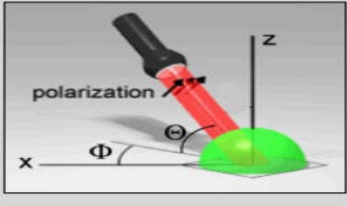
Sweep Parameter: Angle phi (deg)

Minimum: 0

Maximum: 90

Number of Points: 3

Light Source



Absorption

Simple Q Dot - Absolute Fermi Level: ☐ no

Simple Q Dot - Rel. or Abs. Fermi Level: 0eV

Multi-Layer - Fermi Level: 0.7eV

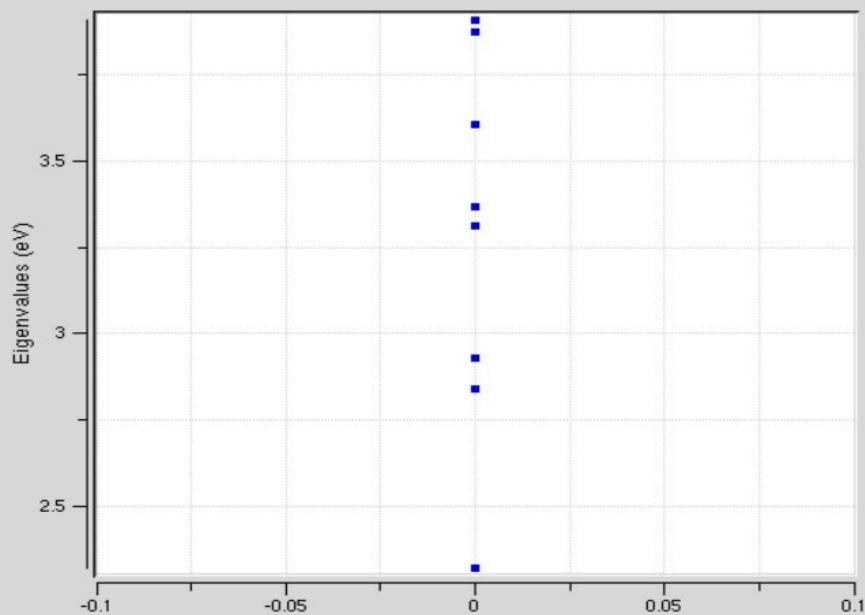
Temperature: ☐ 300K

State Broadening: 0.001

< Structure

Advanced >

Result: All Eigen Energies



1 result

Clear

< Advanced

Case Study 2: Simulate the energy states of cylindrical quantum dots of X=15 nm, Y = 10.5nm and Z = 8 nm.

Quantum Dot Lab
✕ Terminate
➡ Keep for later

1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate

Quantum Dot Structure

Type of Quantum Dot Structure: Simple Quantum Dot

Simple Quantum Dot Options

Shape: Cylinder

Number of States: 8

X Dimensions: 10nm

Y Dimensions: Input field for the length of the dot in the x-direction.

Z Dimensions: Enter a number between 1 and 50 with units of length (A,bohr,in,m)

Lattice Constant: 0.5nm

Effective Mass: 0.067

Energy gap: 1.43eV

< Introduction
Optical >

Quantum Dot Lab
✕ Terminate
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Light Incident Angles

Angle Theta: 45deg

Angle Phi: 0deg

Sweep

Sweep Parameter: Angle phi (deg)

Minimum: 0

Maximum: 90

Number of Points: 3

Light Source

Absorption

Simple Q Dot - Absolute Fermi Level: no

Simple Q Dot - Rel. or Abs. Fermi Level: 0eV

Multi-Layer - Fermi Level: 0.7eV

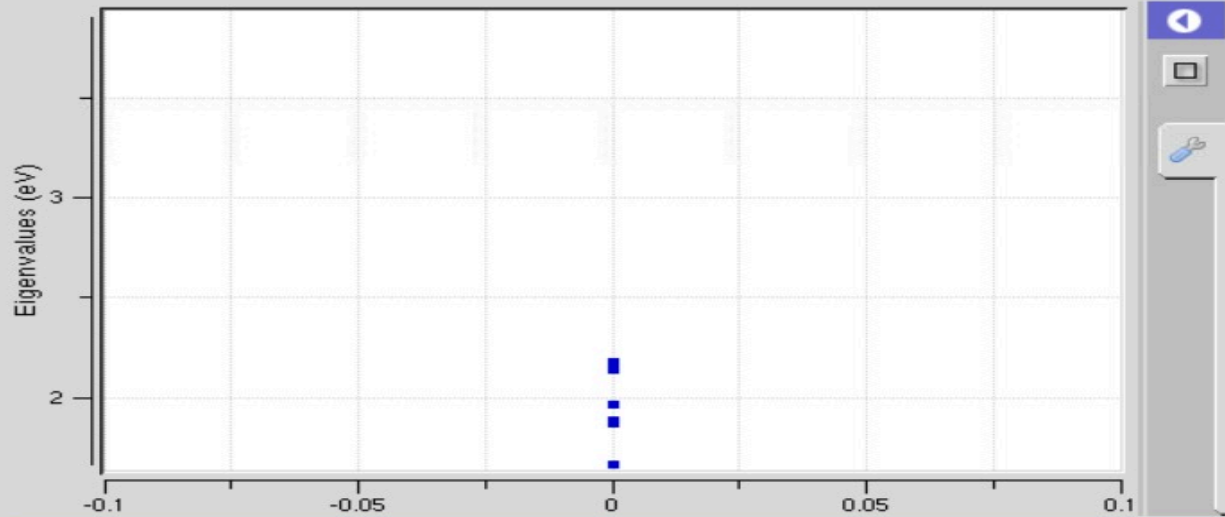
Temperature: 300K

State Broadening: 0.001

< Structure
Advanced >



Result: All Eigen Energies



2 results

Clear One

Clear All

All

► Simulation = #2

Shape = Cylinder
Z Dimensions = 8nm

< Advanced

Case Study 3: Simulate the energy states of two layer cuboid quantum dots

Quantum Dot Lab ✕ Terminate ➡ Keep for later

1 Introduction → 2 **Structure** → 3 Optical → 4 Advanced → 5 Simulate

Quantum Dot Structure

Type of Quantum Dot Structure: Multi-Layer Quantum Dot

Multi-Layer Quantum Dot Options

Shape: Cuboid

Cond. Band States: 6

Val. Band States: 8

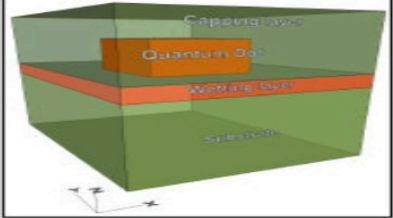
Basis: TB10 band 5 orb sp3s_so

Materials: GaAs - InAs - GaAs

Materials: GaAs - InAs - GaAs

Strain: ☒ yes

Local Bands: ☒ yes



Quantum Dot Alloy Lateral Domain Vertical Layers

X Dimension: 8nm

Y Dimension: 8.5nm

Z Dimension: 5nm

< Introduction

Optical >

Quantum Dot Lab ✕ Terminate ➡ Keep for later

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Light Incident Angles

Angle Theta: 45deg

Angle Phi: 0deg

Sweep

Sweep Parameter: Angle phi (deg)

Minimum: 0

Maximum: 90

Number of Points: 3



Light Source

polarization

x

z

Phi

Theta

Absorption

Simple Q Dot - Absolute Fermi Level: ☒ no

Simple Q Dot - Rel. or Abs. Fermi Level: 0eV

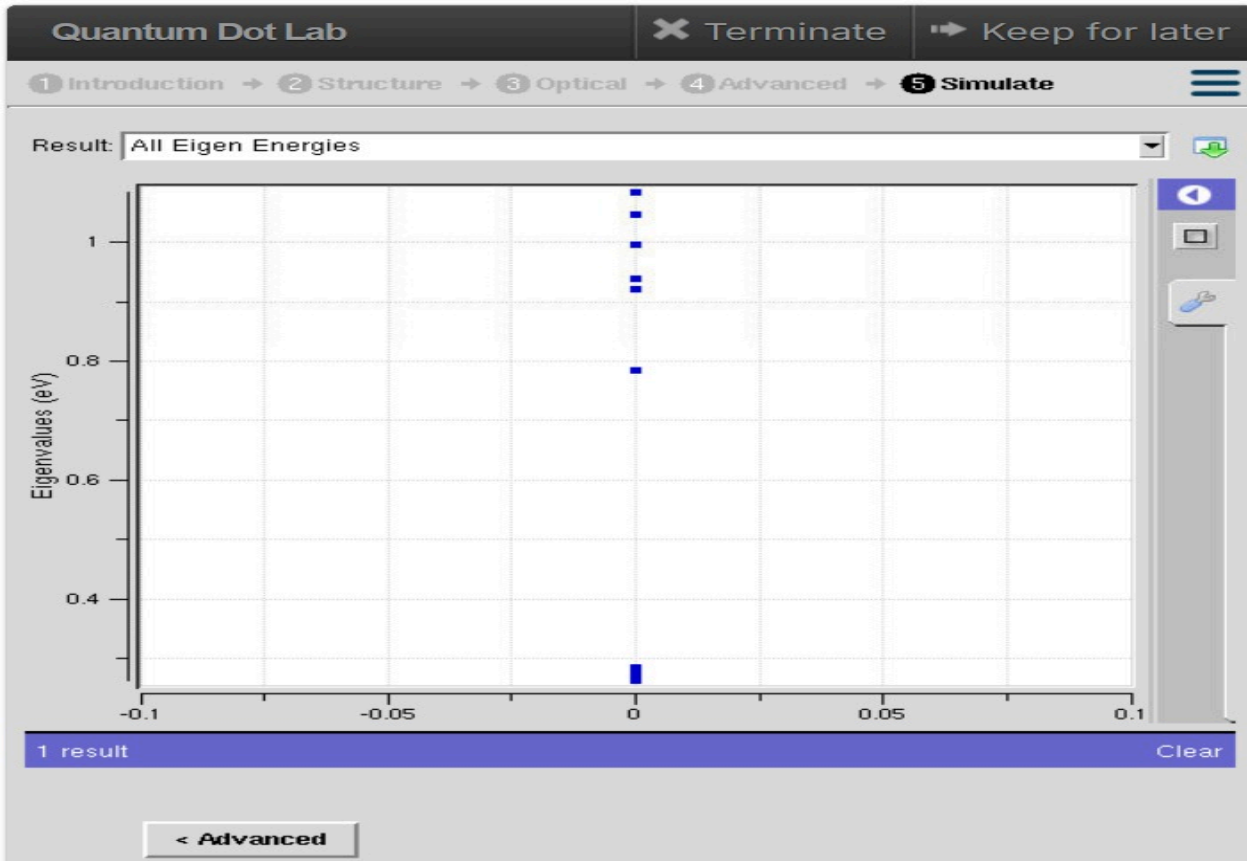
Multi-Layer - Fermi Level: 0.7eV

Temperature: 300K

State Broadening: 0.001

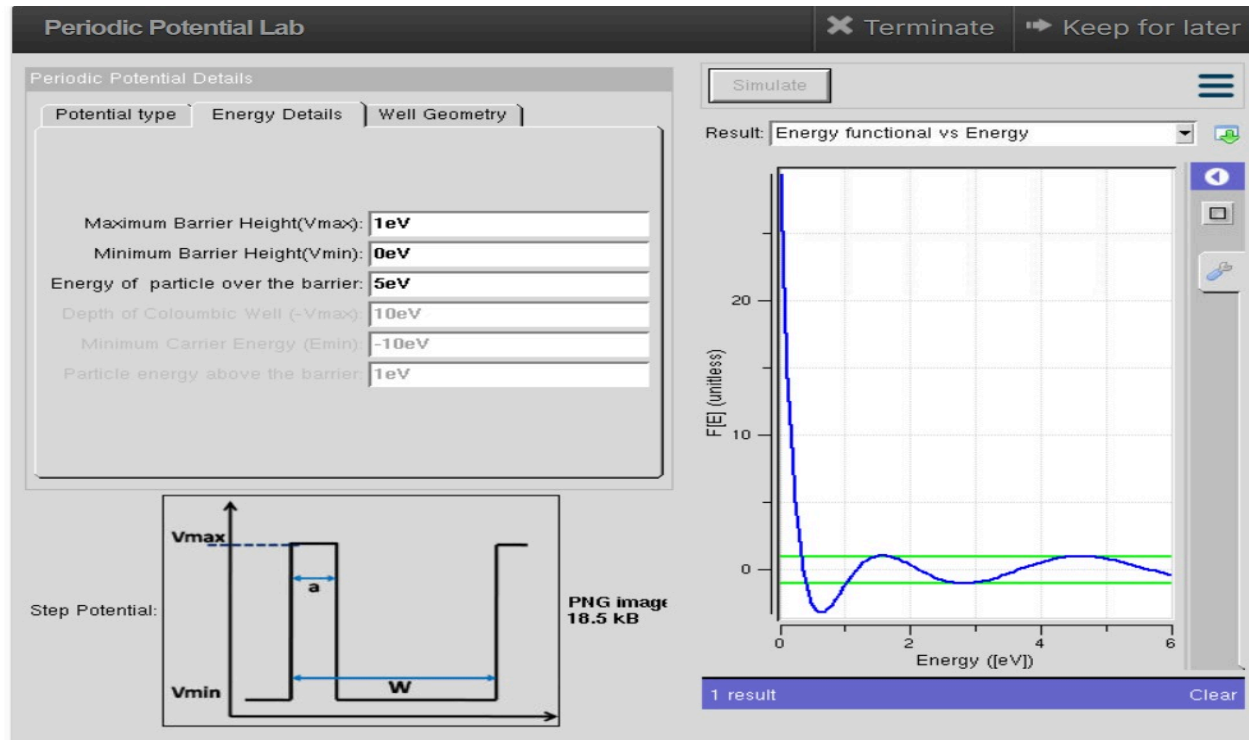
< Structure

Advanced >

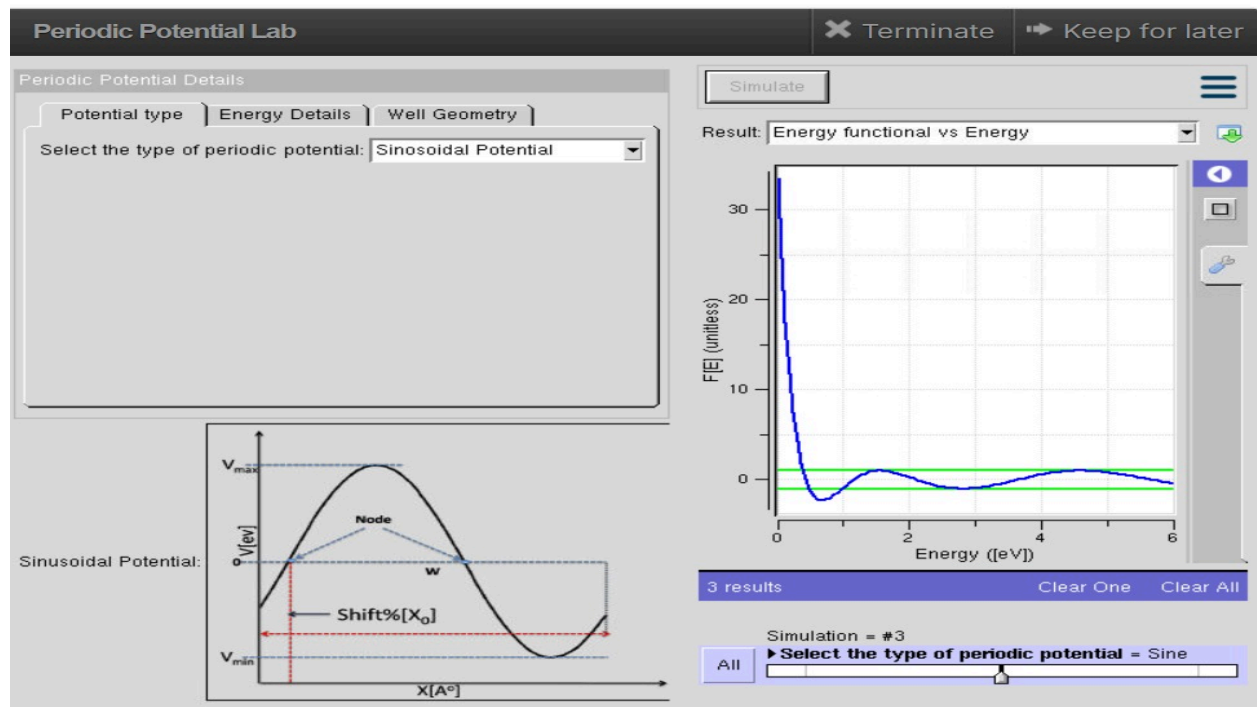


Exp. 4. Kronig Penney Model

Case Study 1: Simulate the output for Rectangular well



Case Study 2: Simulate the output for Sinusoidal Potential



Case Study 3: Simulate the output for Triangular well

